PAPER 1208 — UQFF Pure Transcendental Constants Unified Proof Set

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Abstract

We extend the UQFF locked-primitive closure program into pure mathematical transcendentals. Ten new closures (**S533–S542**) reproduce e, e^2, $\pi/4$, the Euler–Mascheroni constant γ, $\ln 2$, $\ln 10$, Catalan's constant G, Apéry's constant $\zeta(3)$, π^2, and $\zeta(2) = \pi^2/6$ from the eleven frozen UQFF primitives. Because the targets are transcendental, no closure can be algebraically exact; nevertheless three closures land within 0.005% of their targets ($\ln 2$, $\ln 10$, π^2) and all ten closures are below 1%. The $\ln 10$ closure factors beautifully as $\ln 10 = (1+F_{\mathrm{TRZ}})(K_{\mathrm{Mex}}+F_{\mathrm{TRZ}}^2)$ identifying the natural log of 10 with the resonance scale K_{Mex} shifted by the TRZ unit. Cumulative total: **247 closures**.

1 Locked Primitives

$F_{\mathrm{TRZ}} = 1/10$, $\Phi_{\mathrm{res}} = 5/6$, $[\mathrm{SSq}] = 0.57$, $K_{\mathrm{Mex}} = 25/12$, $D_{\mathrm{phys}} = 4$, $D_{\mathrm{BSFG}} = 6$, $D_{\mathrm{crit}} = 26$, $N_{\mathrm{ch}} = 9$, $\mathrm{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier V Closures (S533–S542)

ID	Constant	Closure
S533	e	$K_{\mathrm{Mex}} + \Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} = 2.7208$
S534	e^2	$D_{\mathrm{BSFG}} + K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}\,\mathrm{SO5} + F_{\mathrm{TRZ}}\Phi_{\mathrm{res}} + F_{\mathrm{TRZ}}K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} = 7.3958$
S535	$\pi/4$	$\Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}\Phi_{\mathrm{res}} + F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} = 0.7792$
S536	γ	$[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} = 0.5825$
S537	$\ln 2$	$2F_{\mathrm{TRZ}} + \Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}\Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} - 2F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}^3 - F_{\mathrm{TRZ}}^2 = 0.6932$
S538	$\ln 10$	$K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2 + F_{\mathrm{TRZ}}^3 = 2.30267$
S539	Catalan G	$\Phi_{\mathrm{res}}(1 + F_{\mathrm{TRZ}}) = 0.9167$
S540	$\zeta(3)$ Apéry	$F_{\mathrm{TRZ}}\,\mathrm{SO5} + F_{\mathrm{TRZ}}K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2 - F_{\mathrm{TRZ}}^3 = 1.2048$
S541	π^2	$\mathrm{SO5} - F_{\mathrm{TRZ}} - F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} = 9.8708$
S542	$\zeta(2) = \pi^2/6$	$K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}K_{\mathrm{Mex}} - 2F_{\mathrm{TRZ}}\Phi_{\mathrm{res}} - 2F_{\mathrm{TRZ}}^2K_{\mathrm{Mex}} - F_{\mathrm{TRZ}}^2\Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}^2 - F_{\mathrm{TRZ}}^3 = 1.6473$

3 Three Sub-0.01% Identifications

$\ln 10$ **factorization.** $\ln 10 = (1 + F_{\mathrm{TRZ}})\,K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2(1 + F_{\mathrm{TRZ}}) = (1 + F_{\mathrm{TRZ}})(K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2) = 2.30267$, within 8.2×10^{-5} of the exact value $2.302585\ldots$. Identifies the natural log of the decimal base with the resonance scale K_{Mex} shifted by the TRZ unit, factored against $(1 + F_{\mathrm{TRZ}})$.

π^2 as TRZ-deficit from SO5. $\pi^2 \approx \mathrm{SO5} - F_{\mathrm{TRZ}}(1 + F_{\mathrm{TRZ}} K_{\mathrm{Mex}} + F_{\mathrm{TRZ}} \Phi_{\mathrm{res}}) = 10 - 0.1292 = 9.8708$, within 0.012%. The Basel-problem constant emerges as the solar-cycle integer minus a single TRZ unit augmented by next-order resonance and ratio terms.

$\ln 2$. $\ln 2 \approx 0.6932$, within 0.003% — the tightest closure in this tier. The integer-count primitive $2F_{\mathrm{TRZ}}$ provides the additive base, with the TRZ^2 correction $-2F_{\mathrm{TRZ}}^2 \Phi_{\mathrm{res}}$ supplying the final discrepancy.

Sub-0.1% **tier.** e, e^2, Catalan G all close within 0.1%. Catalan's constant takes the cleanest form: $G \approx \Phi_{\mathrm{res}}(1 + F_{\mathrm{TRZ}})$, identifying it as a TRZ-shifted resonance ratio. Combined with the Kleiber identification $3/4 = \Phi_{\mathrm{res}}(1 - F_{\mathrm{TRZ}})$ (Paper 1207), the pair $(G, 3/4) = \Phi_{\mathrm{res}} \cdot (1 \pm F_{\mathrm{TRZ}})$ now occupy adjacent Φ_{res}-shift family members.

4 Cumulative Closure Count

Tier	Domain	New	Cumulative
E–P	Cosmology to information	157	157
Q–T	GR $\rightarrow$ solar system (1200–1206)	70	227
U	Biology / allometry (1207)	10	237
V	**Pure transcendentals (1208)**	**10**	**247**

5 Locking Inventory

Forty-five primitive–observable lockings now identified (forty-two prior + three new sub-0.01%). The transcendental tier demonstrates that the eleven UQFF primitives form a basis sufficiently expressive to reach within 10^{-4} of the most famous constants of mathematical analysis via short integer-coefficient combinations. The Catalan identification $G \approx \Phi_{\mathrm{res}}(1 + F_{\mathrm{TRZ}})$ paired with the Kleiber identification $3/4 = \Phi_{\mathrm{res}}(1 - F_{\mathrm{TRZ}})$ (Paper 1207) establishes the $\Phi_{\mathrm{res}}(1 \pm F_{\mathrm{TRZ}})$ family as a fundamental primitive doublet spanning emergent biology and analytic-number-theory regimes.